



Team 1 Project:

Parking Lot System



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Feature 1 – Entry/Exit Management and Manual Space Allocation

Duration: 1 week

Goal: Ensure accurate control of vehicle entry and exit operations by allowing operators to manually register movements, validate license plates, and maintain real-time parking spot availability.

Tasks	Description
Vehicle Type Identification	Identify whether the vehicle is a resident, visitor, or authorized guest before registering entry.
Time Stamp Verification	Automatically verify and store accurate system timestamps for all entry and exit records.
Operator Confirmation Step	Require manual confirmation from the operator before saving any entry or exit movement.
Duplicate Plate Detection	Detect and prevent duplicate active entries for the same license plate.
Spot Availability Validation	Verify that the selected parking spot is available before assigning it to a vehicle.
Manual Spot Reassignment	Allow the operator to reassign a parking spot if an error was made during registration.
Error Handling and Alerts	Display clear error messages when invalid data or actions are detected.
Entry/Exit Record Editing	Allow authorized users to correct entry or exit records in case of human error.
System Activity Logging	Keep a log of all manual operations performed by operators for audit purposes.



Feature 2 – Resident Rental and Subscription Management

Duration: 1 week

Goal: Manage resident information, vehicle assignments, and rental payment status to ensure authorized and up-to-date use of parking spaces.

Tasks	Description
Resident Registration	Register new residents into the system with complete personal and apartment information.
Resident Profile Update	Allow editing of resident information such as contact details or apartment number.
Multiple Vehicle Management	Enable residents to register and manage more than one vehicle under their account.
Payment Date Recording	Store the exact payment or renewal date associated with each rental transaction.
Expiration Date Calculation	Automatically calculate subscription expiration dates based on selected rental duration.
Rental Status Validation	Prevent parking access for residents with canceled or expired subscriptions.
Subscription History Tracking	Maintain a historical record of previous rental periods and renewals.
Account Deactivation	Allow administrators to deactivate resident accounts when necessary.
Data Consistency Checks	Ensure all resident, vehicle, and rental data remain synchronized across the system.